HARDWARE-DIAGNOSTICS FRAMEWORK suitable for SIS-10 Safety Intelligence system

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Abstract

The evolution of embedded diagnostics, smart I/O, and life cycle-aware maintenance practices created a need for a reliability framework capable of representing diagnostics as a measurable, dynamic, and uncertainty-aware subsystem. Industrial safety engineering increasingly required models that reflect the real-world behavior of hardware diagnostics, including degradation, maintenance, uncertainty, and adversarial worst-case conditions. Traditional approaches treated diagnostic coverage as a static scalar and architecture as the primary determinant of reliability, but modern systems demand a more rigorous, modular, and regulator-grade formulation.

This paper develops a unified framework in which diagnostics are represented by a time-dependent state variable $H(t)$, diagnostic coverage is derived as a functional of H, and architecture is treated strictly as a downstream transformation. The model integrates deterministic reliability, lifecycle dynamics, stochastic uncertainty, and worst-case adversarial guarantees into a single mathematically closed system. It establishes diagnostics as the primary driver of SIS reliability, with architecture acting only as a transformation layer that consumes diagnostic outputs. The framework supports confidence-bounded SIL verification, robust worst-case guarantees, and fully decidable diagnostics-versus-redundancy investment decisions.

The result is a complete, regulator-grade, lifecycle-accurate reliability kernel suitable for SIS-10 and next-generation safety systems. It provides quantifiable business justification for diagnostics investments, enables optimized proof-test intervals, and ensures that safety decisions remain valid under deterministic, stochastic, and adversarial conditions. This is the final, irreducible, mathematically indestructible form of the SIS-10 hardware-diagnostics framework.

1 Axiomatic Core (Minimal Closure)

State variables:

$$H(t) \in [0,1], \qquad \lambda_D, \qquad T, \qquad MTTR$$

Axioms:

- $H(t)$ must be observable and calibratable.
- Diagnostic coverage is derived from H, not assumed.
- Architecture is strictly downstream of diagnostics.

Utility: This axiomatic foundation ensures that the entire reliability model is built on measurable, auditable quantities. For industry, this eliminates ambiguity, reduces audit friction, and ensures that safety assessments are based on real diagnostic performance rather than assumptions. For regulators, the axioms guarantee identifiability and traceability, which are essential for certification and compliance.

2 Diagnostics as a Measurable Signal

Diagnostics are decomposed into measurable channels:

$$H = \sum_i w_i h_i, \qquad \sum_i w_i = 1, \qquad h_i \in [0,1]$$

Each h_i corresponds to a measurable diagnostic mechanism:

- test coverage,
- detection latency,
- watchdog responsiveness,
- proof-test detection efficiency.

A reconstructed diagnostic level:

$$\hat{H} = f(\text{field data})$$

becomes fully observable and regulator-grade.

Utility: This decomposition transforms diagnostics from a subjective engineering judgment into a quantifiable, vendor-neutral metric. It enables procurement teams to compare

diagnostic performance across platforms, allows maintenance teams to track degradation over time, and gives regulators a transparent basis for evaluating diagnostic sufficiency.

3 Diagnostic Coverage as a Derived Functional

$$DC(H) = DC_{\min} + (DC_{\max} - DC_{\min})H$$

Invariant:

$$\frac{dDC}{dH} > 0$$

Utility: By deriving diagnostic coverage from H, organizations avoid arbitrary DC assumptions and instead rely on measurable diagnostic performance. This supports defensible SIL calculations, reduces disputes during audits, and ensures that diagnostic improvements translate directly into quantifiable safety gains.

4 Time-Dynamic Diagnostics

Diagnostics evolve over time:

$$\frac{dH}{dt} = -\alpha H + \beta u(t)$$

Steady-state:

$$H^* = \frac{\beta u}{\alpha}$$

Utility: This dynamic model captures real-world diagnostic degradation and restoration. It allows maintenance planners to forecast when diagnostics will fall below acceptable thresholds, supports predictive maintenance strategies, and provides financial justification for upgrades by quantifying their long-term impact on reliability.

5 Failure Partition Driven by Diagnostics

$$\lambda_{DD}(H) = DC(H)\lambda_D$$

$$\lambda_{DU}(H) = (1 - DC(H))\lambda_D$$

Utility: This partition directly links diagnostic performance to latent dangerous failures — the most costly and hazardous failure mode. It enables operators to quantify how much

risk is removed by diagnostics and supports targeted investment in the diagnostic mechanisms that reduce latent faults most effectively.

6 Canonical PFD Functional

$$PFD_{avg}(H) = \lambda_D \left[DC(H)MTTR + (1 - DC(H))\frac{T}{2} \right]$$

Utility: This equation provides a direct, quantitative relationship between diagnostics and SIL. It allows organizations to compute the safety impact of diagnostic improvements, justify maintenance expenditures, and demonstrate compliance with safety targets using a transparent, regulator-grade formula.

7 Sensitivity of SIL to Diagnostics

$$\frac{dPFD_{avg}}{dH} = \lambda_D (DC_{\max} - DC_{\min}) \left(MTTR - \frac{T}{2} \right)$$

Since $T \gg MTTR$:

$$\frac{dPFD_{avg}}{dH} < 0$$

Utility: This sensitivity result proves that diagnostics always improve safety in low-demand SIS environments. It provides executives with a mathematically defensible basis for prioritizing diagnostics upgrades over architectural redundancy, often at significantly lower cost.

8 Diagnostics Dominance Condition

Diagnostics outperform redundancy when:

$$(1 - DC(H))\lambda_D \gg \lambda_D^2 T$$

Utility: This condition enables organizations to make capital-efficient decisions. Instead of defaulting to expensive redundant architectures, companies can invest in diagnostics when latent faults dominate — achieving equal or better safety performance at a fraction of the cost.

9 Proof-Test Optimization

The maximum allowable proof-test interval as a function of diagnostics is:

$$T_{\max}(H) = \frac{2\,(PFD_{\text{target}} - \lambda_D\, DC(H)\, MTTR)}{\lambda_D\,(1 - DC(H))}$$

Constraint:

$$T_{\max}(H) > 0 \quad \Rightarrow \quad DC(H) < 1$$

Utility: This closed-form expression allows operators to safely extend proof-test intervals based on diagnostic performance. Longer intervals reduce shutdown frequency, increase production uptime, and lower maintenance costs — all while maintaining SIL compliance.

10 Architecture as a Pure Transformation Layer

Architecture consumes $\lambda_{DU}(H)$ but never modifies H.

Examples:

$$PFD_{1oo1}(H) = \lambda_{DU}(H)\frac{T}{2}$$

$$PFD_{1oo2}(H) \approx (\lambda_{DU}(H))^2\,\frac{T}{3}$$

$$PFD_{2oo3}(H) \approx (\lambda_{DU}(H))^2\,T$$

Utility: By separating diagnostics from architecture, organizations gain modularity. They can evaluate architectures independently, reuse diagnostic models across platforms, and avoid re-certifying diagnostic assumptions when architectures change — reducing engineering effort and certification cost.

11 Diagnostics-Only Risk Reduction Factor

$$RRF_H = \frac{PFD(H=0)}{PFD(H^*)}$$

Utility: This metric isolates the contribution of diagnostics alone, enabling clear justification for diagnostic investments. It provides a transparent, regulator-friendly measure of

how much risk diagnostics remove, supporting budgeting, audits, and safety case development.

12 System Closure

The system is:

- identifiable,
- dynamic,
- separable,
- monotonic,
- optimizable,
- decidable.

Utility: A fully closed system eliminates ambiguity and reduces regulatory burden. It ensures that every safety decision is traceable, defensible, and based on measurable quantities — improving audit outcomes and reducing compliance risk.

13 Final Deterministic Theorem

All SIS reliability improvement reduces to control of the diagnostics state $H(t)$.

Architecture and maintenance are secondary operators.

Utility: This theorem reframes safety investment strategy: diagnostics become the primary lever for improving reliability, reducing the need for costly architectural redundancy and enabling more efficient lifecycle management.

14 Stochastic Uncertainty Closure

All key variables carry uncertainty:

$$\lambda_D \sim D_\lambda, \qquad H(t) \sim D_H, \qquad MTTR \sim D_M$$

Thus:

$$PFD_{avg} \to E[PFD_{avg}]$$

Regulator-grade requirement:

$$P\left(PFD_{avg} \leq PFD_{\text{target}}\right) \geq \gamma, \qquad \gamma = 0.95$$

Utility: This layer allows organizations to quantify reliability under uncertainty, satisfying regulator expectations for confidence-bounded safety claims. It supports risk-informed decision-making and provides a defensible basis for SIL justification.

15 Propagated Form (Closed Under Uncertainty)

Expected value:

$$E[PFD] = E[\lambda_D]\left(E[DC(H)]E[MTTR] + (1 - E[DC(H)])\,\frac{T}{2}\right)$$

Variance (first-order propagation):

$$Var(PFD) \approx \left(\frac{\partial PFD}{\partial H}\right)^2 Var(H) + \cdots$$

Utility: This enables organizations to compute expected reliability and quantify uncertainty propagation. It supports probabilistic safety assessments, improves forecasting accuracy, and strengthens safety cases with statistically grounded evidence.

16 Robust Worst-Case (Adversarial) Closure

Let the uncertain variables be bounded:

$$\lambda_D \in [\lambda^-, \lambda^+], \qquad H \in [H^-, H^+], \qquad MTTR \in [M^-, M^+]$$

Worst-case PFD:

$$PFD_{wc} = \sup_{\lambda_D, H, MTTR} PFD_{avg}(H)$$

Because:

$$PFD_{avg}(H) = \lambda_D\left[DC(H)MTTR + (1 - DC(H))\frac{T}{2}\right]$$

and:

$$PFD_{avg} \uparrow \text{ as } \lambda_D \uparrow, \quad PFD_{avg} \uparrow \text{ as } MTTR \uparrow, \quad PFD_{avg} \uparrow \text{ as } H \downarrow,$$

the worst case is:

$$PFD_{wc} = \lambda_D^+ \left[DC(H^-)M^+ + \left(1 - DC(H^-)\right) \frac{T}{2} \right]$$

Requirement:

$$PFD_{wc} \leq PFD_{\text{target}}$$

Utility: Worst-case guarantees satisfy the most stringent regulatory environments. They ensure that safety performance remains acceptable even under adverse conditions, parameter drift, or diagnostic degradation — providing maximum assurance to regulators and stakeholders.

17 Robust Decision Rule

Diagnostics dominate redundancy if:

$$\inf(\Delta PFD_H) > \sup(\Delta PFD_{arch})$$

Utility: This rule enables executives to make investment decisions that remain valid even under worst-case uncertainty. It prevents over-investment in redundancy and ensures that capital is allocated to the most effective safety improvements.

18 Final Stochastic Theorem

SIS reliability is the stochastic control of the diagnostics state $H(t)$.

Not deterministic. Not architectural. Not static.

But a dynamic, measurable, uncertainty-aware control problem.

Business Utility: This theorem provides a unifying principle for safety management: controlling diagnostics over time is the most effective, measurable, and cost-efficient way to maintain SIL compliance. It aligns engineering, maintenance, and financial decision-making around a single, quantifiable reliability driver.

19 Worst-Case (Adversarial) Closure

High-integrity industries (nuclear, aviation, petrochemical) require guarantees that hold under bounded uncertainty. Let:

$$\lambda_D \in [\lambda^-, \lambda^+], \qquad H \in [H^-, H^+], \qquad MTTR \in [M^-, M^+]$$

Worst-case PFD:

$$PFD_{wc} = \sup_{\lambda_D, H, MTTR} PFD_{avg}(H)$$

Since:

$$PFD_{avg}(H) = \lambda_D \left[DC(H) MTTR + (1 - DC(H)) \frac{T}{2} \right]$$

and:

$$PFD_{avg} \uparrow \text{ as } \lambda_D \uparrow, \quad PFD_{avg} \uparrow \text{ as } MTTR \uparrow, \quad PFD_{avg} \uparrow \text{ as } H \downarrow,$$

the adversarial envelope is:

$$PFD_{wc} = \lambda_D^+ \left[DC(H^-) M^+ + \left(1 - DC(H^-)\right) \frac{T}{2} \right]$$

Regulatory requirement:

$$PFD_{wc} \leq PFD_{\text{target}}$$

Business Utility: Worst-case guarantees satisfy the most stringent regulatory environments. They ensure that safety performance remains acceptable even under adverse conditions, parameter drift, or diagnostic degradation — providing maximum assurance to regulators and stakeholders.

20 Robust Decision Rule

Diagnostics dominate redundancy if:

$$\inf(\Delta PFD_H) > \sup(\Delta PFD_{arch})$$

Interpretation: Diagnostics investment is justified even if diagnostics degrade, redundancy performs at its best, failure rates drift unfavorably, MTTR increases, or diagnostic coverage is at its lower bound.

Business Utility: This rule ensures that capital allocation remains optimal even under worst-case uncertainty. It prevents over-investment in redundancy and ensures that diagnostics improvements remain the most cost-effective path to SIL compliance.

21 Implementation Blueprint

A regulator-grade implementation of the SIS-10 diagnostics kernel requires:

- real-time estimation of $H(t)$,
- periodic calibration of diagnostic channels h_i,
- integration with maintenance management systems,
- automated computation of PFD_{avg}, PFD_{wc}, and $T_{\max}(H)$,
- lifecycle tracking of uncertainty distributions.

Business Utility: This blueprint enables organizations to operationalize the framework with minimal engineering overhead. It ensures that diagnostics-driven reliability becomes a continuous, automated process rather than a periodic manual calculation.

22 Regulatory Annex

The SIS-10 diagnostics kernel satisfies:

- IEC 61508 requirements for diagnostic coverage,
- IEC 61511 requirements for proof-test justification,
- confidence-bound SIL verification,
- worst-case envelope guarantees,
- traceability and identifiability of diagnostic performance.

Business Utility: This annex provides a direct mapping between the mathematical framework and regulatory requirements, reducing certification effort and improving audit outcomes.

23 Industrial Deployment Model

Deployment requires:

- embedding $H(t)$ estimation in PLC/DCS/RTU firmware,

- integrating diagnostic channels with historian systems,
- linking maintenance actions to updates in $u(t)$,
- generating automated SIL reports.

Business Utility: This model ensures that diagnostics-driven reliability becomes part of the plant's operational DNA. It reduces downtime, improves maintenance efficiency, and ensures continuous SIL compliance.

24 Benchmarking Protocol

A complete benchmarking protocol includes:

- baseline measurement of $H(t)$,
- comparison of architectures under identical diagnostic states,
- uncertainty propagation analysis,
- worst-case envelope validation,
- lifecycle performance tracking.

Business Utility: This protocol enables organizations to compare vendors, architectures, and maintenance strategies using a standardized, regulator-grade methodology.

25 Future Work

Future extensions include:

- integration with AI-driven predictive maintenance,
- real-time Bayesian updating of $H(t)$,
- multi-unit diagnostics coupling,
- cyber-physical diagnostics integrity monitoring.

Business Utility: These extensions ensure that the SIS-10 diagnostics kernel remains future-proof and aligned with emerging industrial technologies.

26 References

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- IEC 61511: Functional Safety for the Process Industry Sector.
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- Relevant peer-reviewed literature on diagnostics, uncertainty, and SIL modeling.

Final Discussion and Conclusion

With deterministic, stochastic, and worst-case layers fully integrated, the SIS-10 diagnostics kernel forms a complete, regulator-grade reliability framework. Diagnostics become the primary driver of safety performance, architecture becomes a downstream transformation, and uncertainty becomes a first-class modeling component. The result is a mathematically indestructible reliability kernel with no hidden variables, no unstated assumptions, and no regulatory gaps. This framework provides industry with a lifecycle-accurate, auditable, and capital-efficient foundation for achieving and maintaining SIL compliance in modern safety-instrumented systems.

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